Mathematics

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1 Fundamentals of Higher Mathematics

Generally covers four areas: single-variable calculus, linear algebra, multivariable calculus, and differential equations. As this is fundamentals, the focus is generally on computation instead of proofs.

1.1 Single Variable Calculus

Calculus is the branch of mathematics that studies change and accumulation. It has two main branches:

i. Differential Calculus: Concerned with the rate at which things change (e.g. velocity).

ii. Integral Calculus: Concerned with accumulation (e.g. area under a curve).

Calculus underpins physics, engineering, economics, biology, and more.

Definition 1.1.1. The *derivative* is defined as:

$$\lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0)$$

Definition 1.1.2. A function $f(x)$ is continuous at x_0 when

$$\lim_{x \to x_0} f(x) = f(x_0)$$

Definition 1.1.3. The *right-hand* and *left-hand* limits are as follows:

i. *Right-Hand Limit*: $\lim_{x \to x_0^+} f(x)$ means $\lim_{x \to x_0} f(x)$ for $x > x_0$

ii. *Left-Hand Limit*: $\lim_{x \to x_0^-} f(x)$ means $\lim_{x \to x_0} f(x)$ for $x < x_0$

Definition 1.1.4. A *removable discontinuity* is where

$$\lim_{x \to x_0^+} f(x) = \lim_{x \to x_0^-} f(x)$$

but not at $f(x_0)$, or if $f(x_0)$ is undefined.

Definition 1.1.5. A *jump discontinuity* is where $\lim_{x \to x_0^+} f(x)$ and $\lim_{x \to x_0^-} f(x)$ exists, but not equal.

Definition 1.1.6. A *infinite discontinuity* is where $\lim_{x \to x_0^+} f(x) \neq \lim_{x \to x_0^-} f(x) \in \{+\infty, -\infty\}$

Theorem 1.1.7. *Differentiability Implies Continuity*
If f is differentiable at x_0, then f is continuous at x_0.

Proof.

$$\lim_{x \to x_0} (f(x) - f(x_0)) = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} \right] (x - x_0) = f'(x_0) \cdot 0 = 0$$

□

Definition 1.1.8. *Product Formula*
The derivative of product of two functions.

$$(u \cdot v)' = u' \cdot v + u \cdot v'$$

Definition 1.1.9. *Quotient Formula*
The derivate of ratio of two differentiable functions. Let $v(x) \neq 0$, then

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Definition 1.1.10. *Chain Rule*
The derivative of composition of two functions.

$$\frac{d}{dt} f(g(t)) = f'(g(t))g'(t)$$

May be used to differentiate implicitly defined functions.

Theorem 1.1.11. *Taylor's Theorem*
Let $k \geq 1$ be an integer and let function $f : \mathbb{R} \to \mathbb{R}$ be k-times differentiable at point $a \in \mathbb{R}$. Then there exists a function $h_k : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) = \sum_{i=0}^{k} \frac{f^{(i)}(a)}{i!}(x-a)^i + h_k(x)(x-a)^k$$

where $\lim_{x \to a} h_k(x) = 0$ is the Peano form of the remainder.

Theorem 1.1.12. *Mean Value Theorem*
Let $f : [a, b] \to \mathbb{R}$ be a continuous function on closed interval $[a, b]$ and differentiable on open interval (a, b) where $a < b$. Then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem 1.1.13. *First Fundamental Theorem of Calculus*
Let f be continuous real-valued function on closed interval $[a, b]$ Let F be function defined for all $x \in [a, b]$, by

$$F(x) = \int_a^x f(t)dt$$

Then F is uniformly continuous on $[a, b]$ and differentiable on open interval (a, b), and

$$F'(x) = f(x)$$

for all x in (a, b) so F is an antiderivative of f.

Remark 1.1.14. *Change of Variable*
If $f(x) = g(u(x))$, then $du = u'(x)dx$, hence

$$\int g(u)du = \int g(u(x))u'(x)dx = \int f(x)u'(x)dx$$

For definite integrals,

$$\int_{x_1}^{x_2} f(x)u'(x)dx = \int_{u_1}^{u_2} g(u)d(u)$$

where $u_1 = u(x_1), u_2 = u(x_2)$.

Theorem 1.1.15. *Second Fundamental Theorem of Calculus*
Let $f : \mathbb{R} \to \mathbb{R}$ on $[a, b]$ and F a continuous function on $[a, b]$ which is an antiderivative of f in (a, b),

$$F'(x) = f(x)$$

If f is Riemann integrable on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Remark 1.1.16. *Integration by Trigonometric Substitution*
Evaluate trigonometric integral, work backwards to original variable by using $\text{trig}(\text{trig}^{-1})$ with right-triangle.

i. $a^2 - x^2$ use $x = a \sin u$ and $dx = a \cos u \; du$
ii. $a^2 + x^2$ use $x = a \tan u$ and $dx = a \sec^2 u \; du$
iii. $x^2 - a^2$ use $x = a \sec u$ and $dx = a \sec u \tan u \; du$

Remark 1.1.17. *Systematic Integration of Rational Functions*
For a rational function P/Q (ratio of polynomials), by method of partial fractions, factor Q completely into linear and quadratic factors, and write P/Q as a sum of simpler terms.

i. If $Q(x) = (x-a)^m(x-b)^n$, then the expression is

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_m}{(x-a)^m} + \frac{B_1}{x-b} + \frac{B_2}{(x-b)^2} + \cdots + \frac{B_n}{(x-b)^n}$$

ii. If there are quadratic factor such as $(Ax^2+Bx+C)^p$, then there are terms

$$\frac{a_1x+b_1}{Ax^2+Bx+C}+\frac{a_2x+b_2}{(Ax^2+Bx+C)^2}+\cdots+\frac{a_px+b_p}{(Ax^2+Bx+C)^p}$$

for each factor. To integrate, complete the square and make trigonometric substitution.

Remark 1.1.18. *Integration by Parts*

$$\int_b^a uv'\,dx = uv|_a^b - \int_a^b u'v\,dx$$

This is used when $u'v$ is simpler than uv' (often the case when u' is simpler than u).

Remark 1.1.19. *Arc-Length*
Given $ds=\sqrt{dx^2+dy^2}$, depending on whether to integrate with respect to x,t,y, this is

$$ds=\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx;\quad ds=\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt;\quad ds=\sqrt{\left(\frac{dx}{dy}\right)^2+1}\,dy$$

Remark 1.1.20. *Surface Area for Surface of Revolution*

i. Around x-axis: (requires formula for $y=y(x)$)

$$2\pi y\,ds = 2\pi y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

ii. Around y-axis: (requires formula for $x=x(y)$)

$$2\pi x\,ds = 2\pi x\sqrt{\left(\frac{dx}{dy}\right)^2+1}\,dy$$

Remark 1.1.21. *Polar Coordinates*
Given $x=r\cos\theta$, $y=r\sin\theta$ (or more rarely $r=\sqrt{x^2+y^2}\theta$, $\theta=\tan^{-1}(y/x)$)

i. Find the polar equation for a curve from its equation in (x,y) variables by substitution.
ii. Sketch curves given in polar coordinates and find range of θ for integration.
iii. Integration area formula is then

$$\int_{\theta_1}^{\theta_2}\frac{1}{2}r^2\,d\theta$$

Theorem 1.1.22. *L'Hopital Rule*
For functions f and g defined on open interval I and differentiable on $I\backslash\{c\}$ for a accumulation point c of I, if

$$\lim_{x\to c}f(x)=\lim_{x\to c}g(x)=0\quad or\quad \pm\infty$$

and $g'(x)\neq 0$ for all $x\in I\backslash\{c\}$, and $\lim_{x\to c}\frac{f'(x)}{g'(x)}$ exists, then

$$\lim_{x\to c}\frac{f(x)}{g(x)}=\lim_{x\to c}\frac{f'(x)}{g'(x)}$$

1.2 Linear Algebra

Linear algebra is the branch of mathematics that studies vectors, matrices, and linear transformations. It provides the foundation for solving systems of equations and analysing multidimensional data. Its applications span across machine learning, computer graphics, physics, finance, and engineering.
References from Garcia and Horn (2017) and Axler (2024).

1.2.1 Vector Spaces

Definition 1.2.1. *Vector Space*
Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Then $\mathcal{V}$ is a vector space over a field $\mathbb{F}$ if the following are satisfied:

i. Identities: $\exists\ \mathbf{0} \in \mathcal{V}$ such that $\mathbf{0} + \mathbf{u} = \mathbf{u}\ \forall\ \mathbf{u} \in \mathcal{V}$, and $\exists\ \mathbf{1} \in \mathcal{V}$ such that $\mathbf{1u} = \mathbf{u}\ \forall\ \mathbf{u} \in \mathcal{V}$
ii. Commutativity: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}\ \forall\ \mathbf{u}, \mathbf{v} \in \mathcal{V}$
iii. Associativity: $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ and $(ab)\mathbf{u} = a(b\mathbf{u})\ \forall\ a, b \in \mathbb{F}$ and $\forall\ \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$
iv. Distributive: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ and $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}\ \forall\ a, b \in \mathbb{F}$ and $\forall\ \mathbf{u}, \mathbf{v} \in \mathcal{V}$
v. Additive Inverse: $\forall\ \mathbf{u} \in \mathcal{V}\ \exists$ unique vector $-\mathbf{u} \in \mathcal{V}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
vi. Multiplicative Inverse: $\forall\ \mathbf{u} \in \mathcal{V}$ with $\mathbf{u} \neq \mathbf{0}\ \exists$ unique $\mathbf{w} \in \mathcal{V}$ such that $\mathbf{uw} = 1$

Remark 1.2.2. *Properties of Vector Spaces*
Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$.

i. A vector space has a unique additive identity
ii. Every element in a vector space has a unique additive inverse
iii. $\mathbf{0v} = \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$
iv. $a\mathbf{0} = \mathbf{0}$ for every $a \in \mathbb{F}$
v. $(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$

Definition 1.2.3. *Subspace*
A subspace of an $\mathbb{F}$-vector space $\mathcal{V}$ is a subset $\mathcal{U}$ of $\mathcal{V}$ that is an $\mathbb{F}$-vector space with the same vector addition and scalar multiplication operations as in $\mathcal{V}$. To check $\mathcal{U}$ satisfies the following:

i. Additive Identity: $\mathbf{0} \in \mathcal{U}$
ii. Closure Under Addition: $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ implies $\mathbf{u} + \mathbf{v} \in \mathcal{U}$
iii. Closed Under Scalar Multiplication: $a \in \mathbb{F}$ and $\mathbf{u} \in \mathcal{U}$ implies $a\mathbf{u} \in \mathcal{U}$.

Definition 1.2.4. *Sum of Subspaces*
Suppose $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. The sum of $\mathcal{U}_1, \ldots, \mathcal{U}_m$ as denoted by $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is defined to be set of all possible sums of elements of $\mathcal{U}_1, \ldots, \mathcal{U}_m$:

$$\mathcal{U}_1 + \cdots + \mathcal{U}_m \coloneqq \{\mathbf{u}_1 + \cdots + \mathbf{u}_m : \mathbf{u}_1 \in \mathcal{U}_1, \ldots, \mathbf{u}_m \in \mathcal{U}_m\}$$

Definition 1.2.5. *Direct Sum of Subspaces*
If $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$ such that $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, then every element of $\mathcal{V}$ may be written as

$$\mathbf{u}_1 + \cdots + \mathbf{u}_m$$

where each $\mathbf{u}_j \in \mathcal{U}_j$. Then $\mathcal{V}$ is the direct sum of subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_m$, written as $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$

Definition 1.2.6. *Linear Combination*
Let $\mathcal{U}$ be non-empty subset of $\mathbb{F}$-vector space $\mathcal{V}$. A linear combination of elements of $\mathcal{U}$ is expressed as

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_r\mathbf{v}_r$$

where r is a positive integer, and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r \in \mathcal{U}$, $c_1, c_2, \ldots, c_r \in \mathbb{F}$.
A linear combination is *trivial* if $c_1 = c_2 = \cdots = c_r = 0$.

Definition 1.2.7. *Span*
Let $\mathcal{U}$ be subset of $\mathbb{F}$-vector space $\mathcal{V}$. If $\mathcal{U} \neq \emptyset$, then $\text{span}(\mathcal{U})$ is the set of linear combinations of elements of $\mathcal{U}$. Define $\text{span}(\emptyset) = \{\mathbf{0}\}$. If $\beta = \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ is a list of $r \geq 1$ vectors in $\mathcal{V}$, define $\text{span}(\beta) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$.

References

Axler, S. (2024). *Linear Algebra Done Right* (4 ed.). Undergraduate Texts in Mathematics. Cham: Springer.

Garcia, S. R. and R. A. Horn (2017). *A Second Course in Linear Algebra*. Cambridge: Cambridge University Press.